Impact of Generative AI on Students

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Research Question

Main question

Does the intensity of Generative AI use affect academic performance, learning outcomes, and student well-being?

- Main explanatory variable:
- Source : Kaggle `Weekly_GenAI_Hours`
- Dataset: 50,000 student observations
- Student characteristics
- AI usage patterns
- Academic outcomes
- Well-being indicators

Methodology

We estimate OLS regressions with controls:

- Pre-Semester GPA
- Major Category
- Year of Study
- Traditional Study Hours
- Prompt Engineering Skill
- Tool Diversity
- Paid Subscription
- Institutional Policy

Who use more AI

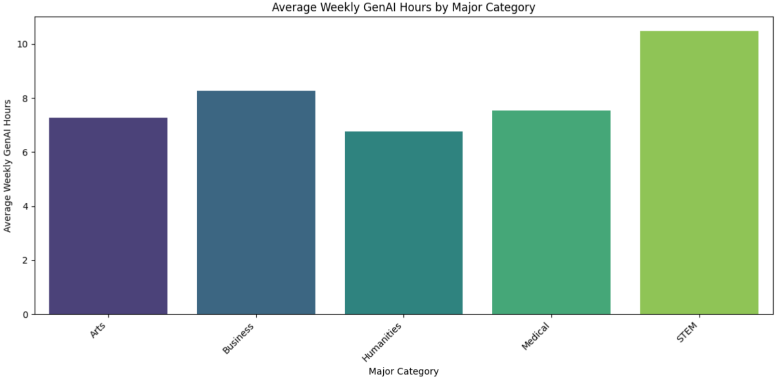

Figure: Use of AI by category

Result 1: Academic Performance

Dependent variable: `Post_Semester_GPA`

- Weekly AI usage is not significantly associated with academic performance.
- The coefficient is close to zero.
- p-value $= 0.535$.

Conclusion

Students who spend more time using AI do not obtain significantly better grades.

Result 2: Skill Retention

Dependent variable: `Skill_Retention_Score`

- AI usage has a negative relationship with knowledge retention.
- Students who use AI more frequently tend to retain fewer skills.

Conclusion

Generative AI use may deteriorate students' skill retention.

Result 3: Anxiety

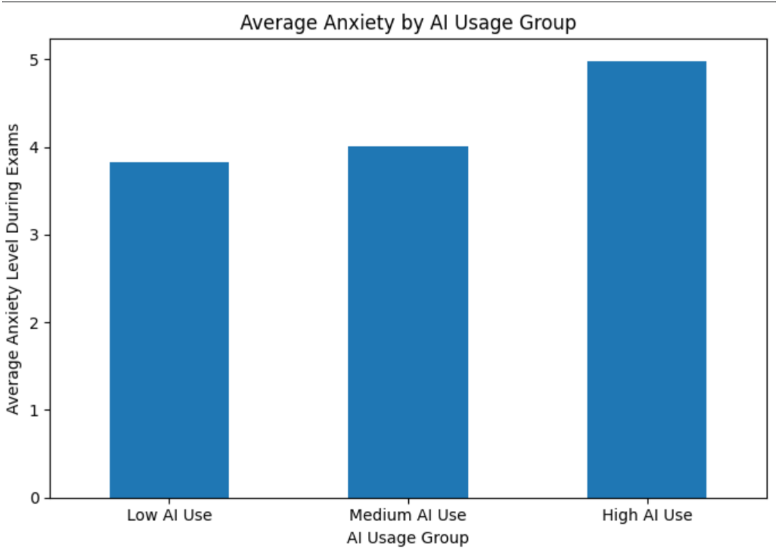

Figure: Average anxiety by AI usage group

Result 3: Anxiety

Dependent variable: `Anxiety_Level_During_Exams`

- Positive and statistically significant relationship.
- Students with higher AI usage report higher anxiety levels.
- Average anxiety increases from 3.82 for low AI users to 4.99 for high AI users.

Conclusion

Heavy AI users tend to experience greater exam anxiety.

Result 4: Burnout Risk

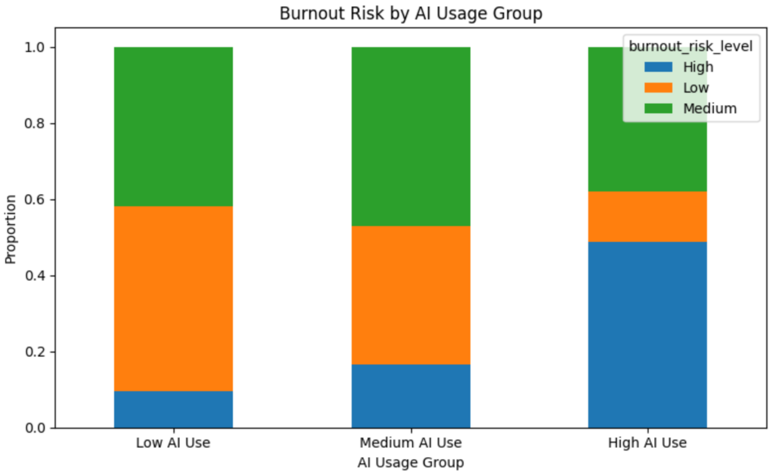

Figure: Burnout risk by AI usage group

Result 4: Burnout Risk

Dependent variable: `Burnout_Risk_Level`

- Strong positive relationship between AI usage and burnout risk.
- `Weekly_GenAI_Hours` coefficient = 0.0418.
- Highly significant: p-value < 0.001.
- High burnout risk rises from 9.4% among low AI users to 48.8% among high AI users.

Main Conclusion

- AI usage does not improve academic performance.
- AI usage deteriorates skill retention.
- AI usage is associated with higher anxiety.
- AI usage is associated with higher burnout risk.

Overall conclusion

Intensive AI use appears to affect student well-being more than academic outcomes.